Particulate Matter
Monitoring & Health Effects

Issue 2026-08-20 | Entry window 20 August 2026 | 18 records in scope, 49 screened out

18 RECORDS IN SCOPE (49 SCREENED OUT)	**14** PARTICLE-SIDE RECORDS (78%)	**1** RECORD WITH A HEALTH ENDPOINT	**0** EFFECT ESTIMATES EXTRACTABLE	**4** METADATA-ONLY DEPOSITS (22%)

Signal of the day

A three-year, ten-node PurpleAir network anchored to collocated reference monitors is now the pollution baseline for a capital city — the distributed low-cost architecture doing the job a regulatory network was never built to do.

Shlipak et al. (*ACS EST Air*) report April 2022–March 2025 of continuous $PM_{2.5}$ from **10 PurpleAir sensors across Addis Ababa**, calibrated against Beta Attenuation Monitors collocated at two contrasting U.S. Embassy sites, plus black carbon from AethLabs MA350 microaethalometers at those same two sites. Network-mean $PM_{2.5}$ was **30 μg/m^3**, exceeding the WHO 24-hour guideline of 15 μg/m^3 on **99% of days**; BC was 3.7 and 7.7 μg/m^3 at the two BC sites, an *intra-urban* factor of two that a single-site regulatory monitor would have erased. Aethalometer-model apportionment put **94% of BC on fossil fuel** and 6% on biomass burning — inverting the assumption that a sub-Saharan African capital's carbon is predominantly biomass. Two design choices are worth stealing: the two-site BAM anchor (rather than one) constrains the calibration across contrasting source regimes, and tying the deployment to the forthcoming MAIA satellite gives the surface network a defined validation role rather than leaving it as a standalone dataset. For distributed low-cost PM work this is the strongest existing-practice reference published this window.

What changed today

- **This is an atmospheric-science issue, not an epidemiology one.** 14 of 18 records measure particles rather than people, and **exactly one** carries a human health endpoint. **No record in the window published a confidence-interval-bearing risk estimate**, so the forest plot is omitted rather than populated with chamber fold-changes and model fit statistics — see *Corpus analytics*.
- **Two independent papers attack the same measurement gap: source resolution without a mass-only monitor.** Shlipak et al. separate fossil from biomass BC by aethalometer model in Addis Ababa; Saarikoski et al. (*ACP*) show at a Helsinki traffic site that relying on hydrocarbon-like OA alone to estimate traffic primary OA **underestimates it by ∼50%**, because a second, more oxygenated traffic-related factor (distinct $C_2H_4O_2^+$, $C_2H_5O_2^+$, $C_3H_5O_2^+$ fractions) also peaks with the morning rush. Any source-apportionment product built on an HOA-only convention inherits that bias.
- **Wildfire smoke is being reframed as a joint-pollutant problem.** Yu et al. (*ES&T*) fit a spatiotemporal Bayesian neural field over 2004–2023 CONUS at 5 km to estimate daily probabilities of *co-occurring* $PM_{2.5}$ and ozone extremes, and find the amplified co-occurrence risk propagating into the Midwest and Northeast on long-range transport. Two monitors, not one, is the implied network requirement.
- **Personal exposure and neighbourhood proxies diverge sharply, and the divergence is quantified.** Zundel et al. (medRxiv preprint) report personal $PM_{2.5}$ of 28.2±20.0 vs 5.1±4.7 μg/m^3 on smoke vs non-smoke days against neighbourhood PurpleAir estimates of 87.6±80.2 vs 12.3±7.0 — personal concentrations were *lower* than the outdoor proxy but rose fivefold, with a within-person slope of only **2.3 μg/m^3 per 10 μg/m^3** of neighbourhood $PM_{2.5}$. Four participants; treat the slope as an order of magnitude, not an estimate.
- **Aerosol phase state is becoming a tractable field measurement.** Do et al. (*ES&T*) measured RH-dependent viscosity of ambient $PM_{2.5}$ by poke-and-flow (6.2×10^4–1.3×10^7 Pa s at 20–40% RH) and propagated it to nighttime N_2O_5 uptake coefficients of 0.01–0.07 — up to an order of magnitude below the liquid-particle assumption of 0.1 that chemical transport models still use.
- **Metadata-only deposits are now a fixed 20–25% tax.** 4 of 18 today (Elsevier *Atmos Environ* ×2, *Atmos Res*, Springer Nature *JESEE*). Semantic Scholar and Europe PMC returned nothing for all four; they are too new. They ship with an explicit METADATA ONLY marker and no inferred quantity.
- **Screening-side note.** 27 of the 49 exclusions came from a single journal (MDPI *Sensors*) whose ISSN was added to the Crossref sweep this run and returned **zero** in-scope records — it indexes sensing hardware of every kind, not airborne particle instrumentation. Recorded for the next sweep-list revision.

Corpus analytics

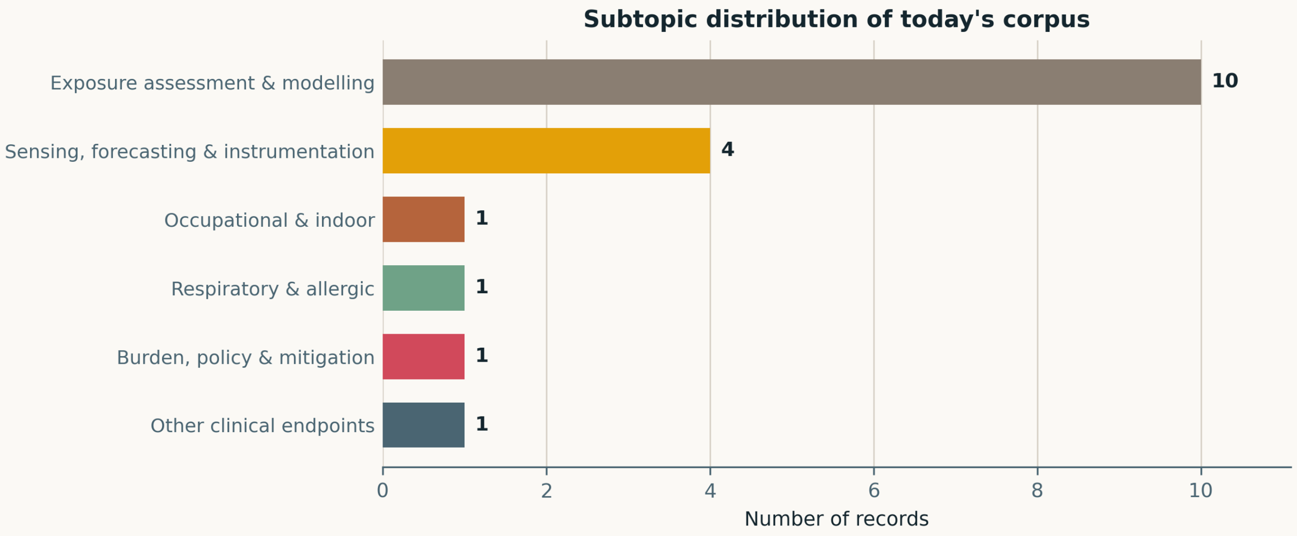
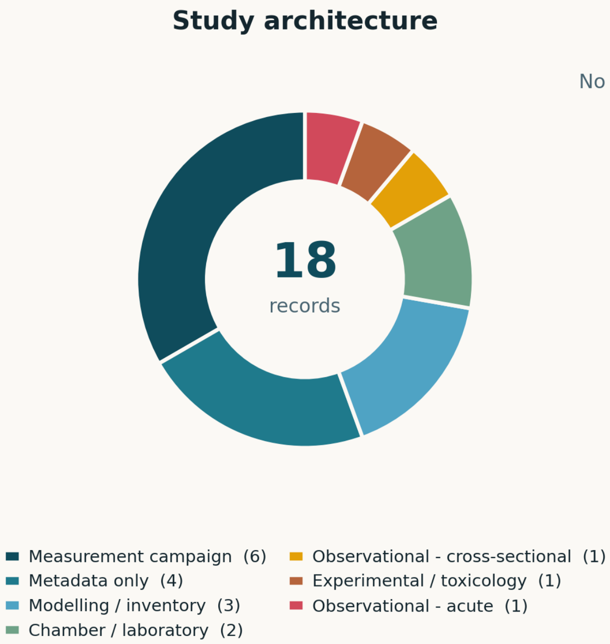


Fig. Subtopic assignment of the 18 in-scope records. Assignment is single-label by dominant contribution. *Exposure assessment & modelling* takes 10 of 18 and *Sensing, forecasting & instrumentation* a further 4; the six clinically oriented clusters between them hold three records. Four of the ten subtopic constants (mechanistic toxicology, cardiovascular/metabolic, neuro/mental health, reproductive/developmental) are empty — an unusually lopsided day, and the direct consequence of the harvest's shape: PubMed's two axes returned 8 unique PMIDs of which 5 shipped, while the Crossref-by-ISSN leg contributed the other 13.



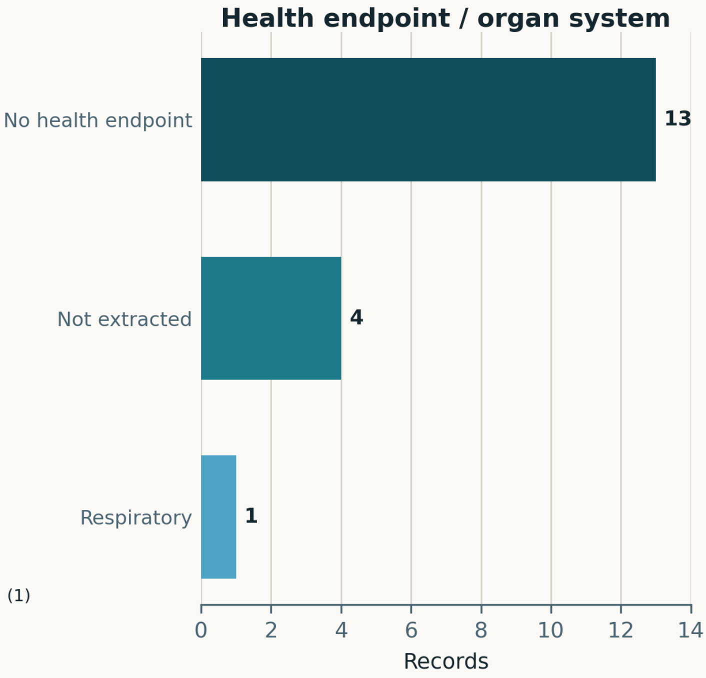


Fig. Left — study architecture. Measurement campaigns lead at 6, followed by metadata-only deposits at 4 and modelling/inventory work at 3; there is no cohort, no case-crossover and no trial in this window. Right — health endpoint by organ system. Thirteen records explicitly measure particles rather than people and four more are metadata-only deposits whose endpoint could not be extracted, leaving **exactly one record with a stated human health endpoint** — Zhang S et al., respiratory, a panel study whose abstract reports direction and significance but no interval estimates.

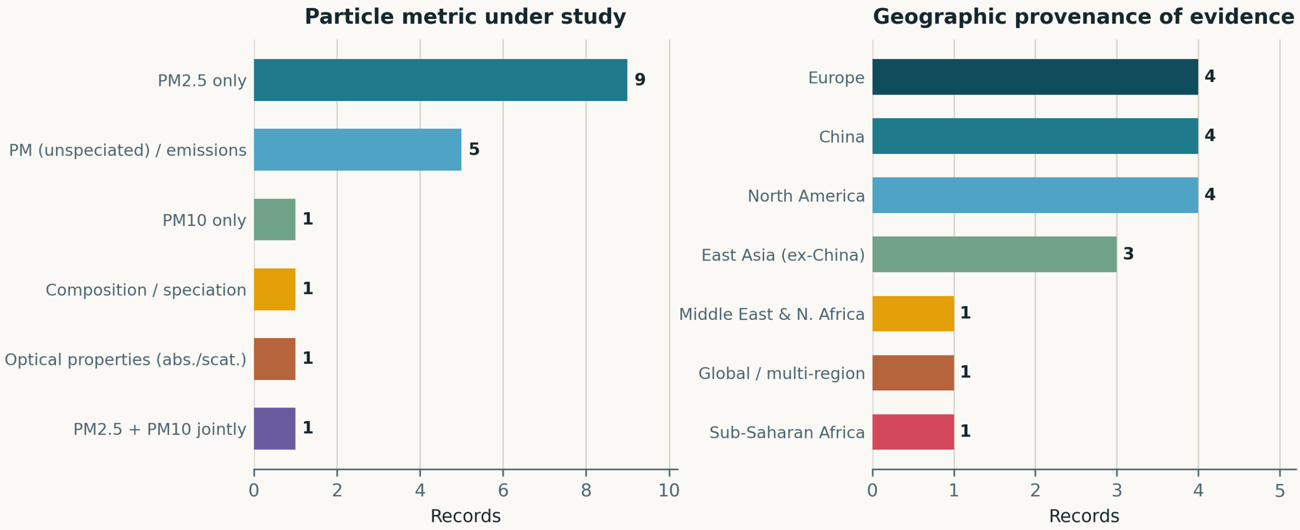


Fig. Left — particle metric. PM$_{2.5}$ alone is the exposure in 9 records; a further 5 concern unspeciated PM or emissions, and one each covers PM$_{10}$ alone, joint PM$_{2.5}$/PM$_{10}$, optical properties (the Riyadh dust AOD/Ångström analysis) and explicit composition/speciation. Right — geographic provenance. Coverage is wider than usual: North America, China and Europe hold 4 each, East Asia excluding China 3, and there is one record each from sub-Saharan Africa (Addis Ababa) and the Middle East (Riyadh). Both of the latter are monitoring-network papers, not health studies.

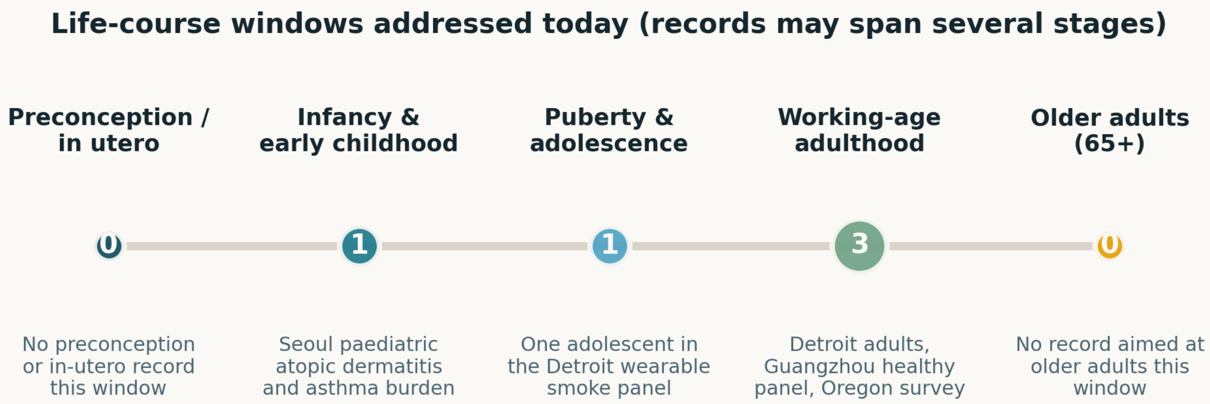


Fig. Life-course windows addressed, counted over the four records that involve human participants at all. Working-age adults account for three (the Detroit wearable panel, the Guangzhou healthy-adult panel, the Oregon resident survey); one paediatric record (Seoul atopic dermatitis and asthma, metadata-only) and one adolescent participant inside the Detroit panel complete the count. **No record this window addressed the preconception or in-utero window, and none targeted adults over 65** — the two stages that ordinarily dominate this figure.

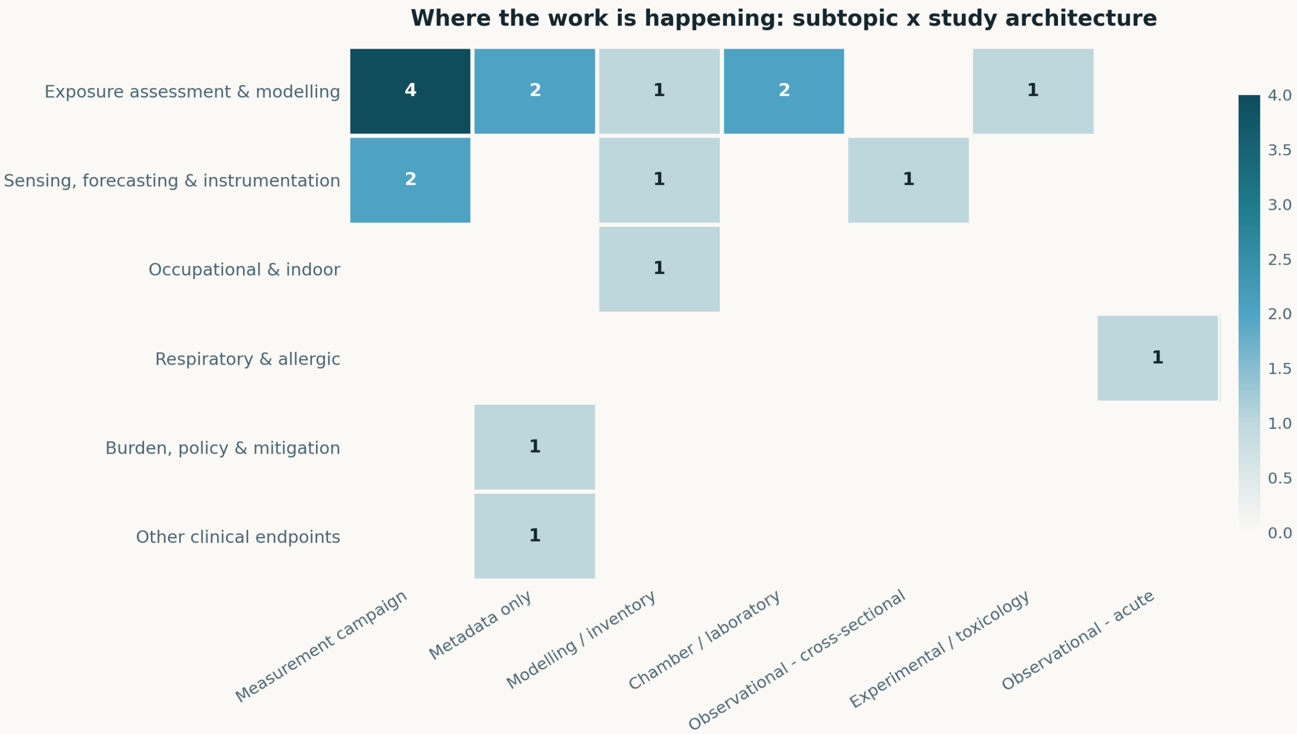


Fig. Subtopic × study-architecture cross-tabulation. The mass sits in a single block — measurement campaigns and modelling within *Exposure assessment* — and the health-side rows are almost empty. Counted against the store — a sensing- or exposure-side record carrying a non-null human health endpoint — that instrumentation/epidemiology bridge appears in **11 of the 19 preceding August issues** and is **absent today**. On an 18-record day dominated by one leg of the harvest, that absence is a sampling artefact of the window, not evidence about the field. (The 19 August issue put this count at 12; recounting under an explicit null-endpoint rule gives 11.)

No forest plot this issue

Every quantity published in this entry window is an absolute concentration or deposition rate (Addis Ababa $PM_{2.5}$ and BC; Swiss microplastic deposition, 881 MPs $m^{-2}\,d^{-1}$, 95% CI 562–1199 at the urban site), a chamber or model fold-change (0.75–1.3× organic-aerosol mass enhancement; ∼40-fold NO_x enhancement under Cl• cycling), or a fit statistic ($R^2 = 0.955$, RMSE 9.05 for the Linfen AQI model). **None of these is a risk estimate**, and pooling them onto the shared logarithmic risk axis would corrupt the weekly rollup, which aggregates this field across issues. `EFFECTS` is therefore empty for 2026-08-20 by decision, not by omission, and figure 5 is not rendered.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation 4 records

Shlipak et al. 2026 ACS EST Air

Spatial and temporal trends of ambient $PM_{2.5}$ and source-apportioned black carbon in Addis Ababa

Three continuous years (Apr 2022–Mar 2025) from the MAIA surface network: 10 PurpleAir units across the city calibrated against collocated Beta Attenuation Monitors at two contrasting U.S. Embassy sites (Central, Jacros), plus AethLabs MA350 microaethalometers at those two sites. Network mean $PM_{2.5}$ **30 μg/m^3**; BC 3.7 and 7.7 μg/m^3; the WHO 24 h guideline of 15 μg/m^3 was exceeded on **99% of days**. Diurnal peaks at morning traffic and under nighttime inversion, strongest in the wet season; highest concentrations west of the city centre. Aethalometer-model apportionment attributes **94% of BC to fossil combustion**, 6% to biomass — the opposite of the usual prior for the region. Directional and back-trajectory analysis raises long-range shipping transport from the Gulf of Aden as a regional contributor. Limitation: only two sites carry a reference anchor and BC, so the calibration transfer across the other eight nodes is assumed rather than demonstrated. The single most directly transferable deployment design in this window.

doi: 10.1021/acsestair.6c00140

Zundel et al. 2026 (preprint) medRxiv

Wearable personal $PM_{2.5}$ monitoring during wildfire smoke episodes in Metro Detroit

AirBeam3 wearables on four participants (one adolescent, three adults) during southeast Michigan smoke episodes, with neighbourhood outdoor $PM_{2.5}$ taken as the mean of the three nearest PurpleAir units and smoke days identified from state advisories. **Personal 28.2±20.0 vs 5.1±4.7 μg/m^3** on smoke vs non-smoke days; neighbourhood **87.6±80.2 vs 12.3±7.0**. Within participants, each 10 μg/m^3 of neighbourhood $PM_{2.5}$ mapped to only **2.3 μg/m^3** personal — personal exposure is both attenuated and poorly tracked by the outdoor proxy, yet still put >25% of participant-days over the EPA 24 h standard. $n = 4$ and no formal uncertainty on the slope: this is a feasibility demonstration, and the attenuation factor should not be reused as a calibration constant. Preprint, not peer reviewed.

doi: 10.64898/2026.07.30.26359253

Clark & Bashar 2026 Risk Anal

How IoT enables the Protective Action Decision Model during a wildfire event

Survey of 1,200 Oregon residents (Jun–Aug 2021) about their responses to the 2020 fire season, framed against the Protective Action Decision Model and the availability of hyper-local real-time $PM_{2.5}$ from IoT devices. The finding that matters for network designers is negative: residents were **overwhelmed by the data and reported inadequate guidance** on how to act on it. Retrospective self-report a year after the event, and the survey does not establish who actually had sensor access — so this constrains interface and messaging design rather than measuring sensor efficacy. Still the rare paper in this watch that treats the last mile of a dense network as the problem.

doi: 10.1111/risa.70319

Wei et al. 2026 iScience

Stacked Transformer-XGBoost model with SHAP interpretability for AQI prediction in Linfen

Transformer temporal encoder stacked with XGBoost through temporally consistent out-of-fold stacking, on 2013–2024 air quality and meteorological series from a heavily industrialised Chinese city. Reported $R^2 = 0.955$**, RMSE 9.05, MAE 2.91**; SHAP attributes the variance to $PM_{2.5}$, NO_2, lagged AQI and seasonal terms. The performance claim is against conventional ML and DL baselines on the same series, with no external site held out, so it is an in-sample-city result. The out-of-fold stacking discipline is the part worth borrowing — naive stacking on autocorrelated air quality series leaks badly.

doi: 10.1016/j.isci.2026.116396

Exposure assessment & modelling 10 records

Yu et al. 2026 Environ Sci Technol

Probabilistic assessment of $PM_{2.5}$ and ozone co-occurring extremes under wildfire smoke

Spatiotemporal Bayesian neural field (ST-BayesNF) integrating satellite smoke-coverage products, ground $PM_{2.5}$ and ozone, and meteorological reanalysis to give daily probabilities of *joint* extremes at 5 km over the contiguous US, 2004–2023. The model discriminates co-extreme from non-co-extreme days and is most sensitive to smoke density interacting with boundary-layer height and near-surface temperature; predicted co-occurrence risk rises during smoke events not only in the western US but increasingly in the Midwest and Northeast under long-range transport. No health outcome is modelled — the health premise (co-occurrence raises risk beyond either pollutant alone) is imported from prior literature and not tested here.

doi: 10.1021/acs.est.6c04403

Saarikoski et al. 2026 Atmos Chem Phys

Role of hydrocarbons and oxygenated organic species in traffic-derived aerosol

Four Aerodyne AMS campaigns (2018–2024) at a Helsinki traffic site with PMF source apportionment. Beyond hydrocarbon-like OA peaking at the morning rush, a second traffic-related factor (TrOA) peaks in the morning but stays elevated longer, with mass spectra between HOA and biomass-burning OA and distinct $C_2H_4O_2^+$ (m/z 60), $C_2H_5O_2^+$ (61) and $C_3H_5O_2^+$ (73) fractions. **Using HOA alone to estimate traffic primary OA underestimates it by ∼50%.** The authors are explicit that TrOA's origin — atmospheric processing versus direct emission from modern vehicles — is unresolved, which is exactly the ambiguity that makes the 50% figure a bound rather than a correction factor. Directly relevant to any speciated or source-resolved sensing product.

doi: 10.5194/acp-26-11817-2026

Do et al. 2026 Environ Sci Technol

Field-constrained diurnal viscosity of urban $PM_{2.5}$ and implications for N_2O_5 uptake

Poke-and-flow measurement with fluid-dynamics inversion on $PM_{2.5}$ collected in Ansan, South Korea (summer 2024): **6.2×10^4–1.3×10^7 Pa s at 20–40% RH**, exceeding 10^8 Pa s below ~10% RH. Pooled with Seoul and Beijing data into a Vogel–Tammann–Fulcher temperature–RH parametrisation and pushed through an ML surrogate over hourly meteorology for 14 megacities. Predicted viscosity is systematically lower at night everywhere (RH plasticisation beating nocturnal cooling), giving nighttime N_2O_5 uptake coefficients of **0.01–0.07** against the conventional liquid assumption of 0.1. Weakness stated by the authors: the parametrisation rests on 24 h composition, so no time-resolved chemistry enters the diurnal prediction it is being used to make.

doi: 10.1021/acs.est.6c04811

Wang Q et al. 2026 Atmos Chem Phys

Quantitative analysis of aerosol acidity and its driving factors in the Guanzhong area

One year of water-soluble inorganic ions in $PM_{2.5}$ on the semi-arid Guanzhong Plain, with thermodynamic pH reconstruction and sensitivity-based driver attribution. Annual mean pH **3.8 ± 1.0** (winter > spring > summer > autumn), shifting from 2–5 on clean days toward 3–6 on polluted days. Temperature contributes 22.3–33.8% of pH variation across all seasons, NH_x 11.4–44.9% (dominant in autumn/winter), sulfate 8.5–10.8%; Ca^{2+} is a spring-specific driver. RH shows a non-monotonic effect with an inflection at 60–85% RH, attributed to deliquescence-driven liquid water content. Attribution rests on a thermodynamic model rather than measured pH, which is the standing caveat for this whole literature.

doi: 10.5194/acp-26-11785-2026

Kommula et al. 2026 Environ Sci Technol

Comparable organic aerosol mass enhancement during photochemical and dark oxidation of biomass burning emissions

Chamber aging of emissions from savannah grass, savannah wood and boreal forest surface fuel under varied combustion conditions, comparing daytime photochemical with nighttime dark oxidation. OA is 82–99% of PM_1 in the fresh emissions. Both aging routes yield **comparable net OA mass**, which PMF traces to more efficient primary-OA loss during daytime offsetting more efficient gas-phase oxidation. The resulting mass enhancement of **0.75–1.3**$\times$ matches field observations and closes a long-standing lab–field discrepancy. Applies to low-NO_x remote conditions; the authors say so, and that is where the proposed model simplification is and is not safe.

doi: 10.1021/acs.est.5c15841

Albugami 2026 Atmosphere

Dust event characteristics, transport pathways and source regions over Riyadh

34 dust events over 2024–2025 identified from AERONET by a dual optical criterion (AOD_{500} $\geq$0.50 and 440–870 nm Ångström exponent $\leq$0.50), cross-checked against NOAA ISD present-weather and visibility and traced with HYSPLIT back-trajectories, PSCF and CWT. **82% fell in March–May**; mean Ångström exponent 0.28, mean fine-mode fraction 0.26. Surface present-weather codes confirmed 26 of 34; the other 8 (24%) were classified as elevated dust invisible at the surface station — a clean demonstration that column and surface instruments disagree by design, not by error. Source–receptor analysis points to the Mesopotamian Basin, with the northern Rub' al-Khali secondary. Single-author, single-station, two years.

doi: 10.3390/atmos17080802

Ashta et al. 2026 Atmos Chem Phys

Wet and dry atmospheric deposition of microplastics across a Swiss site gradient

Four-weekly wet and dry deposition sampling over one year (May 2024–May 2025) at urban (Zurich), suburban (Dübendorf), rural (Magadino, Payerne) and mountainous (Chaumont) sites, with focal-plane-array μ-FTIR over the 20–215 μm range and an explicit measurement-uncertainty treatment. Urban deposition **881 MPs m^{-2} d^{-1} (95% CI 562–1199)** and **53 μg m^{-2} d^{-1} (17–107)**; the other four sites are similar to each other at 249–331 MPs m^{-2} d^{-1} (140–478). Scaled to land use, 219 t or 3.8×10^{14} particles per year below 2000 m nationally, excluding tyre wear. The 20 μm lower cut is the binding limitation: the inhalable fraction sits mostly below it.

doi: 10.5194/acp-26-11803-2026

Wang T et al. 2026 Environ Sci Technol

Chlorine radical cycling on photoactive particulate matter amplifies photolytic NO_x release

Bench photochemistry showing chloride on photoactive PM surfaces oxidised by photogenerated holes to Cl•, which accelerates conversion of nitrite intermediates to gaseous NO_x while chloride is regenerated. ~**40-fold NO_x enhancement** under the model conditions, robust across humidity, chloride salt, substrate (including soot) and nitrogen source (ammonium, 4-nitrophenol). Rate scales with photoactive-matter loading and PM composition. Extrapolated to **0.7–6.2% of NO_2 variability** in the study region — a small number derived from a large laboratory enhancement, and the gap between the two is the result's main uncertainty.

doi: 10.1021/acs.est.6c06768

Naghizade et al. 2026 Atmos Environ

Multi-class organic marker profiling of urban PM_{10} in Strasbourg

METADATA ONLY — seasonal distribution of exhaust, non-exhaust and biomass-burning tracers in northeastern France. No abstract in the Crossref deposit; Semantic Scholar and Europe PMC both returned nothing on 20 August. No quantity is inferred here. Flagged for the next issue that can reach the full record.

doi: 10.1016/j.atmosenv.2026.122306

Chuang et al. 2026 Atmos Res

Simulating $PM_{2.5}$ in southern Taiwan under high-pressure return flow (Kao-Ping Experiment)

METADATA ONLY — a KPEx modelling study of a synoptic regime that traps $PM_{2.5}$ over the Kao-Ping region. No abstract in the deposit and no indexed full text on 20 August. No quantity inferred.

doi: 10.1016/j.atmosres.2026.109286

Respiratory & allergic 1 record

Zhang S et al. 2026 Environ Sci Technol

Respiratory and cardiovascular effects of ozone modified by $PM_{2.5}$

Two complementary designs testing one hypothesis: that 30-day mean $PM_{2.5}$ *modifies* the acute cardiorespiratory response to ozone, and that unmodelled modification is part of why the ozone literature is inconsistent. In a healthy-adult panel, short-term ozone was associated with significantly larger decrements in FEV_1, FEF_{25-75} and PEF, and larger rises in systolic and diastolic blood pressure, under **$PM_{2.5}$ above 22 μg/m^3**. A meta-regression across published studies agreed: study areas with higher annual $PM_{2.5}$ showed larger ozone effects on FEV_1 and blood pressure. **The abstract reports direction and significance but no interval estimates**, which is why this record contributes nothing to the forest plot. Panel size is not stated in the abstract either. Mechanistically the claim is priming rather than co-exposure confounding, and the meta-regression cannot separate the two — areas with high $PM_{2.5}$ differ in many ways. Still the day's one health record, and the modification framing is directly relevant to designing multi-pollutant sensor nodes.

doi: 10.1021/acs.est.6c05604

Occupational & indoor exposure 1 record

Zhao et al. 2026 Indoor Air

CFD-based prediction of air change rate and PM in a dynamically occupied operating room

Class 10,000 operating room in Dalian (Chinese GB 50333-2013), combining field measurement with CFD across 45 orthogonal scenarios, Sobol sensitivity, and three optimisation routes (Bayesian, multi-objective, response surface) to calibrate a non-uniformity coefficient and fit ACH–occupancy–PM. Response surface gave the lowest prediction error (**RMSE 0.032, MAE 0.024**); multi-objective optimisation was the most stable across runs (C_v 0.005). The premise is sound — fixed as-built ACH standards ignore occupancy dynamics — but the model is calibrated on one room in one climate, and the PM metric is cleanroom particle count rather than a health-relevant mass fraction.

doi: 10.1155/ina/5838379

Burden, policy & mitigation 1 record

Anastasopolos et al. 2026 Atmos Environ

Mortality attributed to ambient air pollution by anthropogenic emission sector in Canada

METADATA ONLY — a comparative health-burden analysis run at multiple geographic scales with sectoral attribution. Nine authors, Environment and Climate Change Canada style. No abstract in the Crossref deposit and nothing indexed elsewhere on 20 August; the sector-resolved burden numbers, which are the point of the paper, are therefore not reportable yet. Flagged for a later issue.

doi: 10.1016/j.atmosenv.2026.122310

Other clinical endpoints 1 record

Lee et al. 2026 J Expo Sci Environ Epidemiol

Short-term ambient VOC mixtures, paediatric atopic dermatitis and asthma in Seoul

METADATA ONLY — short-term effects and attributable burden of ambient volatile organic compound *mixtures* on two paediatric outcomes. Included at the edge of scope: the exposure is VOCs rather than particles, but mixture-burden methods on a shared urban monitoring backbone are directly transferable, and the same Seoul network carries $PM_{2.5}$. No abstract available on 20 August, so neither the mixture method nor the burden fraction is reportable.

doi: 10.1038/s41370-026-00965-5

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Zhang S et al.	Respiratory and cardiovascular effects of ozone modified by $PM_{2.5}$	*Environ Sci Technol*	Panel study	30-day mean $PM_{2.5}$ as effect modifier of short-term O3	Guangzhou, China + literature
Occupational & indoor					
Zhao et al.	CFD-based prediction of air change rate and particulate matter in a dynamically occupied operating room	*Indoor Air*	Physical exposure model	Indoor PM (Class 10,000 cleanroom)	Dalian, China
Sensing, forecasting & instrumentation					
Clark & Bashar	How IoT enables the Protective Action Decision Model during a wildfire event	*Risk Anal*	Cross-sectional survey	Hyper-local real-time $PM_{2.5}$ (IoT)	Oregon, USA
Shlipak et al.	Spatial and temporal trends of ambient $PM_{2.5}$ and source-apportioned black carbon in Addis Ababa	*ACS EST Air*	Network deployment + field evaluation	$PM_{2.5}$ + black carbon (BC)	Addis Ababa, Ethiopia
Wei et al.	Stacked Transformer-XGBoost model with SHAP interpretability for air quality index prediction	*iScience*	Machine-learning model	AQI driven by $PM_{2.5}$, PM_{10}, NO2	Linfen, China

First author	Focus	Journal	Design	Metric	Region
Zundel et al. 2026	Wearable personal $PM_{2.5}$ monitoring during wildfire smoke episodes in Metro Detroit	*medRxiv*	Proxy validation vs personal exposure	Personal $PM_{2.5}$ (AirBeam3) vs PurpleAir	Detroit, USA
Exposure assessment & modelling					
Albugami	Dust event characteristics, transport pathways and source regions over Riyadh	*Atmosphere*	Field campaign / remote sensing	Coarse-mode dust (AOD, Angstrom exponent)	Riyadh, Saudi Arabia
Ashta et al.	Wet and dry atmospheric deposition of microplastics at urban, suburban, rural and mountainous sites	*Atmos Chem Phys*	Deposition sampling	Microplastics 20-215 um	Switzerland
Chuang et al.	Simulating $PM_{2.5}$ in southern Taiwan under high-pressure return flow during the Kao-Ping Experiment	*Atmos Res*	Metadata only (no abstract)	$PM_{2.5}$ (not extractable)	Kao-Ping, Taiwan
Do et al.	Field-constrained diurnal viscosity variability of urban $PM_{2.5}$ and implications for N2O5 uptake	*Environ Sci Technol*	Laboratory metrology	$PM_{2.5}$ phase state / viscosity	Ansan, South Korea + 14 megacities
Kommula et al.	Comparable organic aerosol mass enhancement during photochemical and dark oxidation of biomass burning emissions	*Environ Sci Technol*	Chamber emission characterisation	PM1 organic aerosol / BB-SOA	Chamber (Finland)
Naghizade et al.	Multi-class organic marker profiling of urban PM_{10} in Strasbourg	*Atmos Environ*	Metadata only (no abstract)	PM_{10} organic markers (not extractable)	Strasbourg, France
Saarikoski et al.	Role of hydrocarbons and oxygenated organic species in traffic-derived aerosol	*Atmos Chem Phys*	Measurement campaign	Submicron organic aerosol (AMS)	Helsinki, Finland
Wang Q et al.	Quantitative analysis of aerosol acidity and its driving factors in the Guanzhong area	*Atmos Chem Phys*	Measurement campaign	$PM_{2.5}$ water-soluble inorganic ions, pH	Guanzhong Plain, China
Wang T et al.	Chlorine radical cycling on photoactive particulate matter amplifies photolytic NOx release	*Environ Sci Technol*	Controlled chamber factorial	Photoactive PM surfaces, soot	Laboratory
Yu et al.	Probabilistic assessment of $PM_{2.5}$ and ozone co-occurring extremes under wildfire smoke	*Environ Sci Technol*	Bayesian + geospatial regression	$PM_{2.5}$ + O3 joint extremes	Contiguous USA
Burden, policy & mitigation					
Anastasopolos et al.	Comparative health burden analysis at multiple geographic scales: mortality attributed to ambient air pollution by sector	*Atmos Environ*	Metadata only (no abstract)	Sectoral anthropogenic $PM_{2.5}$ (not extractable)	Canada
Other clinical endpoints					
Lee et al.	Short-term effects and attributable burden of ambient VOC mixtures on paediatric atopic dermatitis and asthma	*J Expo Sci Environ Epidemiol*	Metadata only (no abstract)	Ambient VOC mixture (not extractable)	Seoul, South Korea

Provenance and method

Entry window **20 August 2026**, contiguous with the 19 August issue. **PubMed** (E-utilities and connector) was queried on `[Date - Entry]` along a health axis and a monitoring/sensing axis as separate queries: 6 and 3 local hits, 2 and 5 connector hits, **8 unique PMIDs**, of which 5 shipped. **Europe PMC** returned 0 — and the topic-free control query (`CREATION_DATE:[2026-08-20 TO 2026-08-20]`, no topic terms) also returned 0 records of any kind, so this is index lag, not a broken leg. **OpenAlex** returned HTTP 429 for the 18th consecutive run and remains dead. **arXiv** returned nothing inside the 3-day recency screen. **Crossref by ISSN** carried the issue: 5 deposits across the 8 standing journals plus 54 across 18 supplementary ISSNs added this run, of which 13 shipped. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Four records shipped as METADATA ONLY: the publisher deposited no abstract and neither Semantic Scholar nor Europe PMC had indexed them; no quantity is inferred for those records. **No effect estimate with a confidence interval was published in this window**, so the forest plot is omitted — see the note in *Corpus analytics*. 49 records were screened out with a logged reason, 27 of them from one supplementary ISSN (MDPI *Sensors*) that returned no in-scope work. All DOI links resolve to the publisher of record and were checked by `check_dois.py` before this issue was built. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.